ImageNet-R-50: single-adapter Spaced Repetition Training

Results and experimental question

Does confidence-based spaced repetition improve a single continuing rank-16 adapter compared with uniform replay under identical realized exposure? The ImageNet-R split is unchanged: 24,000 training images, 6,000 test images, and fifty four-class tasks in the existing seed-1993 order.

Uniform replay has higher final accuracy and lower final NLL at 2 of the two tested budgets. This comparison tests the selected SRT recipes, not every possible confidence threshold or spacing rule.

At the 1,024-equivalent budget, SRT finishes at 79.267% accuracy and 0.9585 NLL. Relative to its exposure-matched uniform control, the differences are -1.267 accuracy points and +0.0993 NLL. Its accuracy difference from joint rank 16 is +0.400 points.

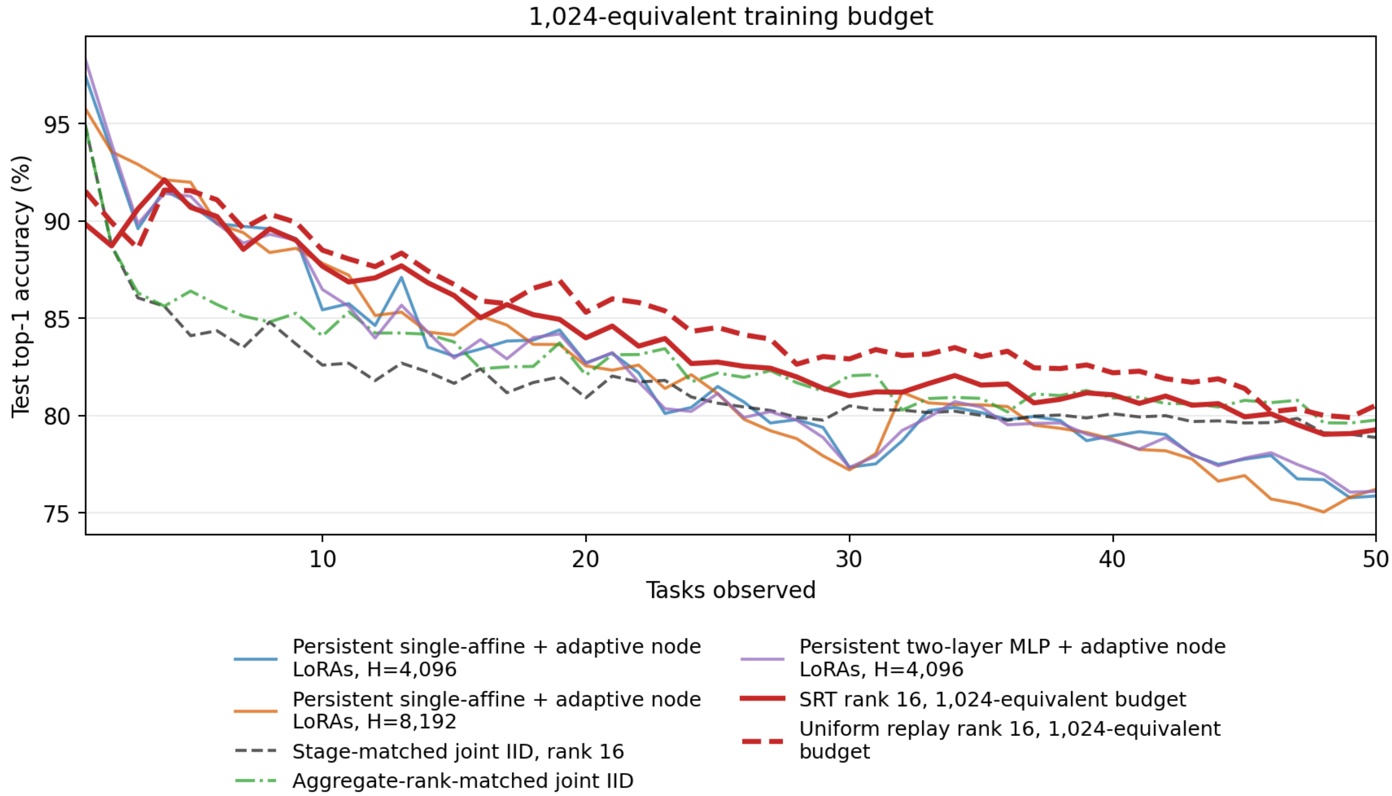
At the 4,096-equivalent budget, SRT finishes at 77.333% accuracy and 1.1100 NLL. Relative to its exposure-matched uniform control, the differences are -3.067 accuracy points and +0.1977 NLL. Its accuracy difference from joint rank 16 is -1.533 points.

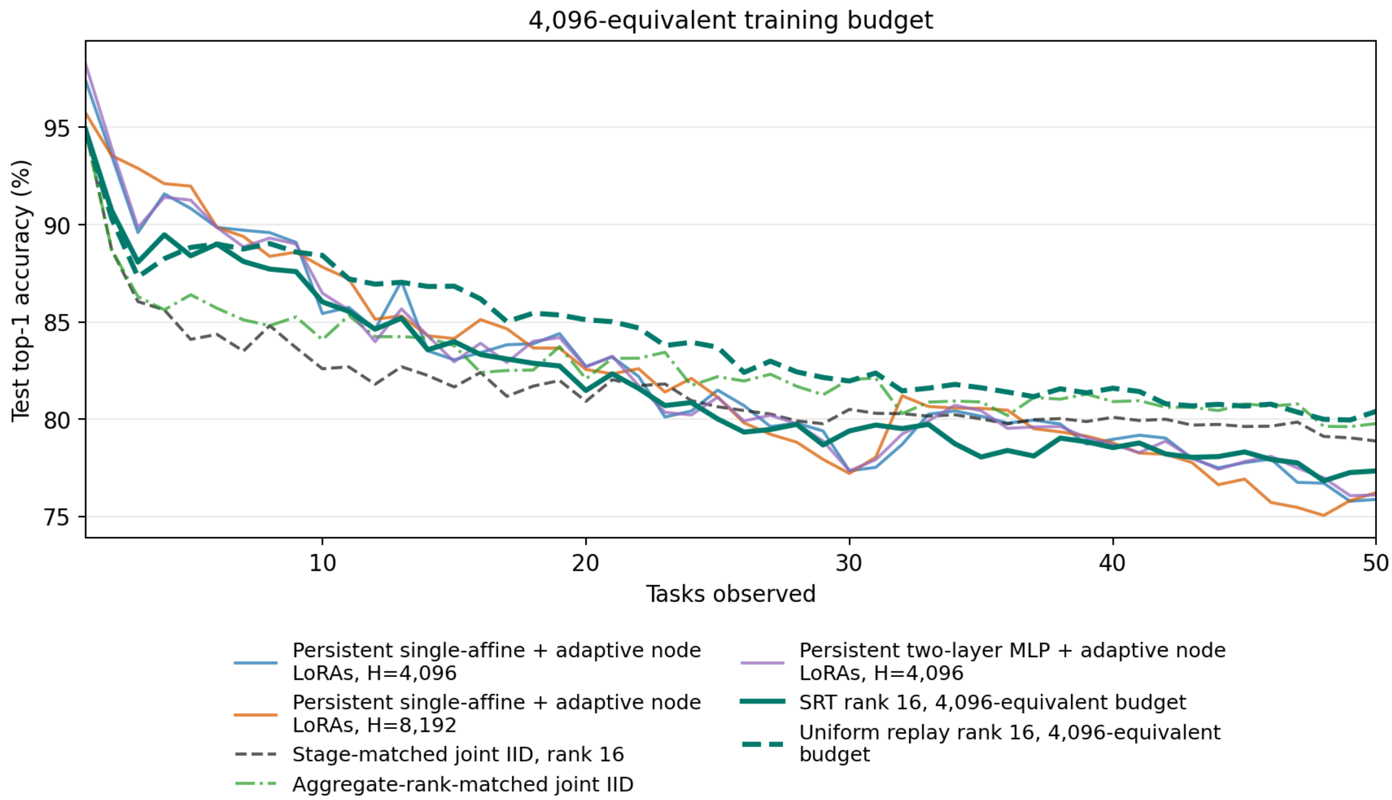
Mean accuracy is the arithmetic mean of the fifty stage test accuracies. NLL is uncalibrated, all-seen-class cross-entropy; lower is better. These are single-seed results, not estimates of training-run variability. Final-run times below exclude calibration.

Condition	Final acc.	Mean acc.	Final NLL	Train min.
SRT rank 16, 1,024-equivalent budget	79.267%	83.916%	0.9585	14.52
Uniform replay rank 16, 1,024-equivalent budget	80.533%	85.059%	0.8592	14.05
SRT rank 16, 4,096-equivalent budget	77.333%	81.908%	1.1100	40.11
Uniform replay rank 16, 4,096-equivalent budget	80.400%	84.197%	0.9123	40.46

Full-stream comparison at both work budgets

Each panel compares one SRT/uniform pair with the five existing task-free conditions. Historical curve names, values, and colors are unchanged. A budget of H-equivalent work means 4 x (current images + min(H, historical images)) presentations per task, not a memory cap. Both methods can revisit every arrived training image.

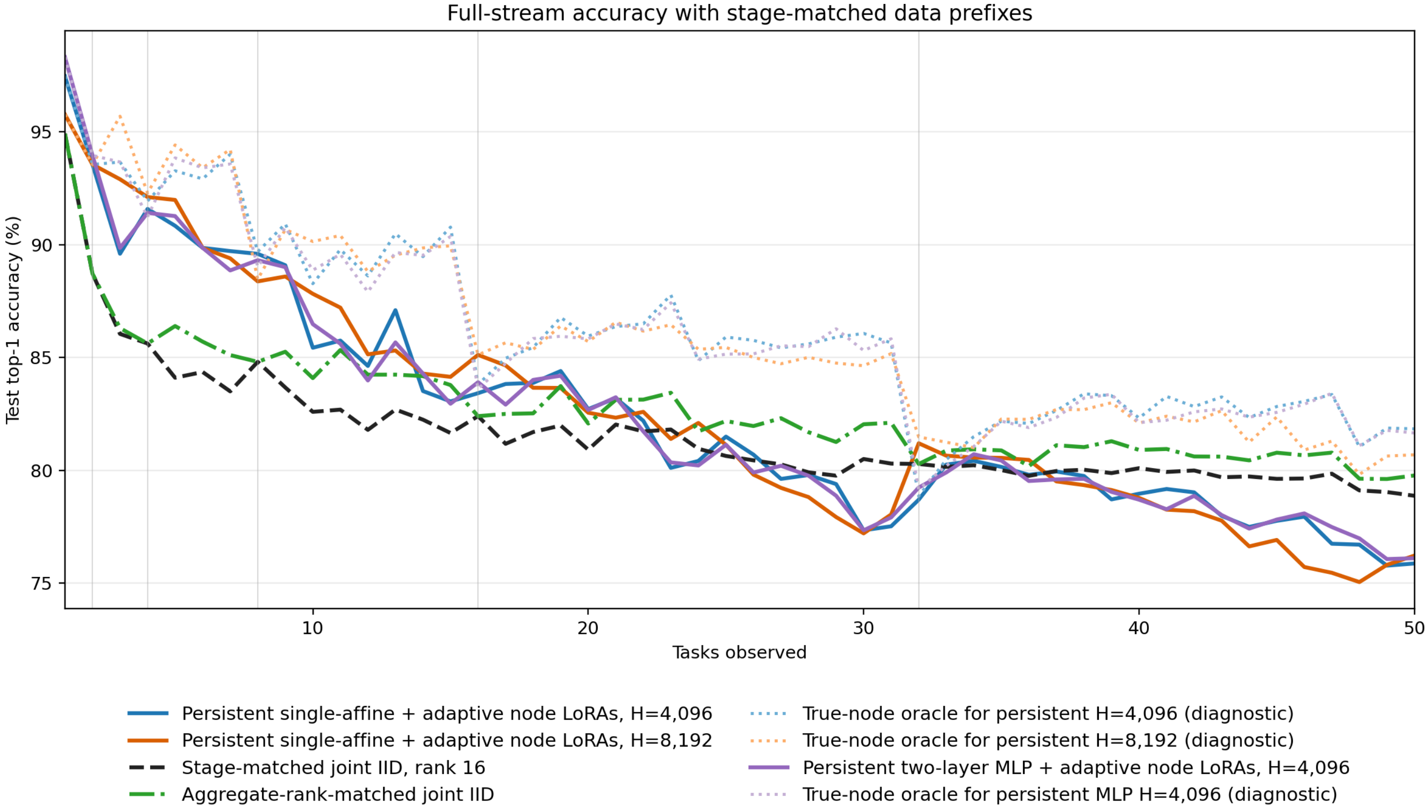




Carried-forward accuracy figure

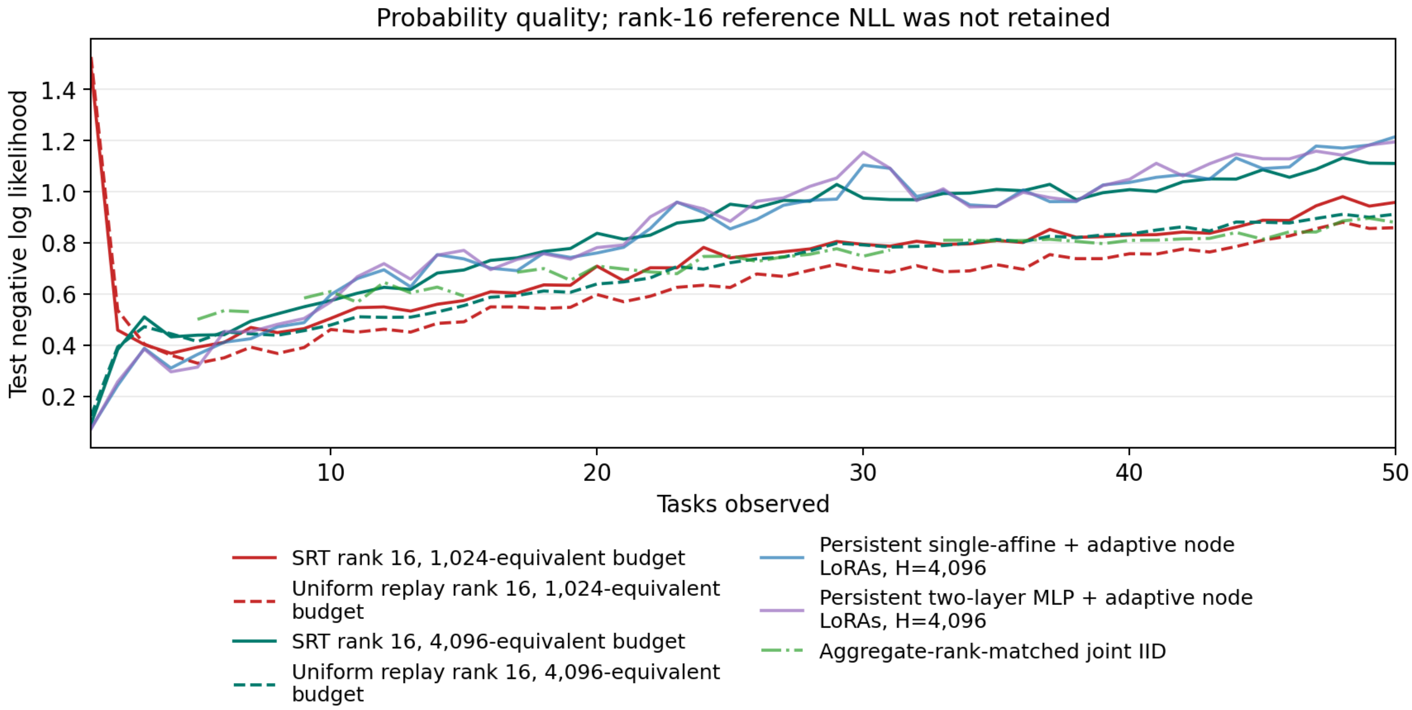
This is the original full-stream accuracy figure, copied byte-for-byte from the previous report. The pale dotted curves are label-aware true-node diagnostics for the frontier models, not deployable methods and not SRT baselines. No previous result or report was rewritten.

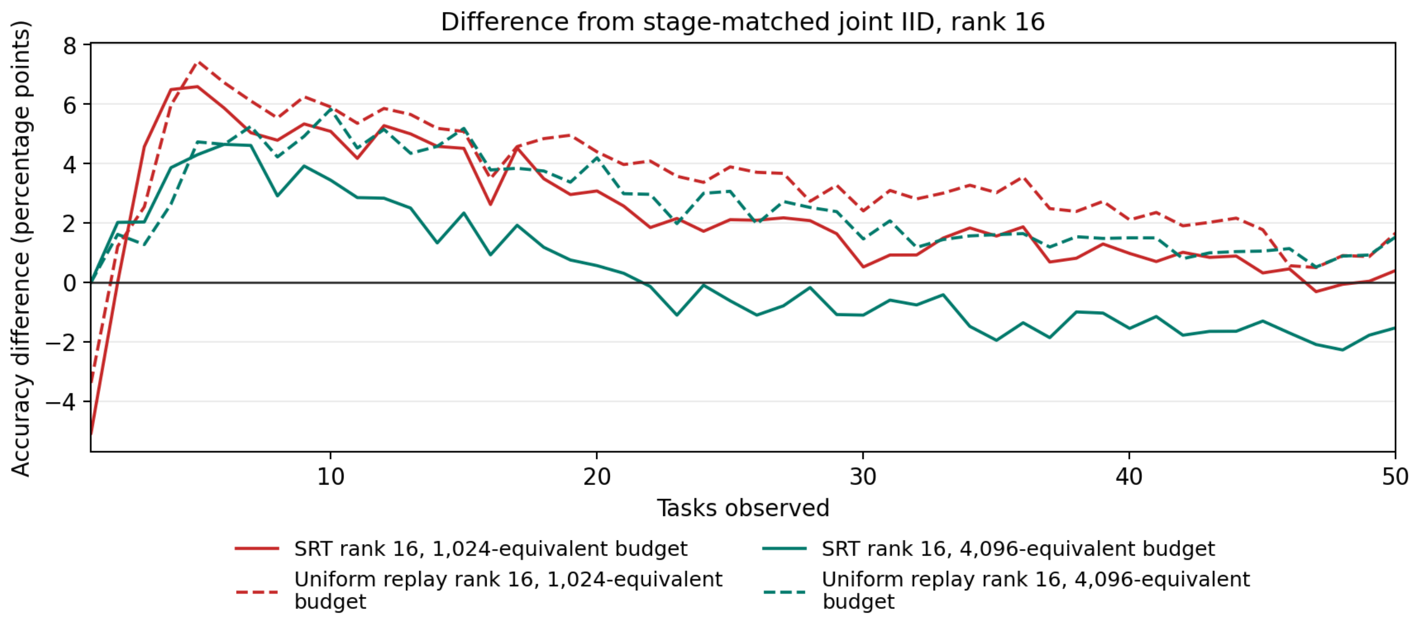
Joint rank 16 uses the same adapter targets, rank, and affine classification architecture as SRT, but retrains on each full prefix. Aggregate-rank-matched joint IID matches the total rank of the frontier nodes, not the rank of the single SRT adapter.



Probability quality and the joint-IID gap

The rank-16 joint source did not retain NLL, so no rank-16 NLL curve is fabricated. The lower panel uses its retained accuracy curve. Neither joint reference is an execution gate or a mathematical upper bound.





Measured training work

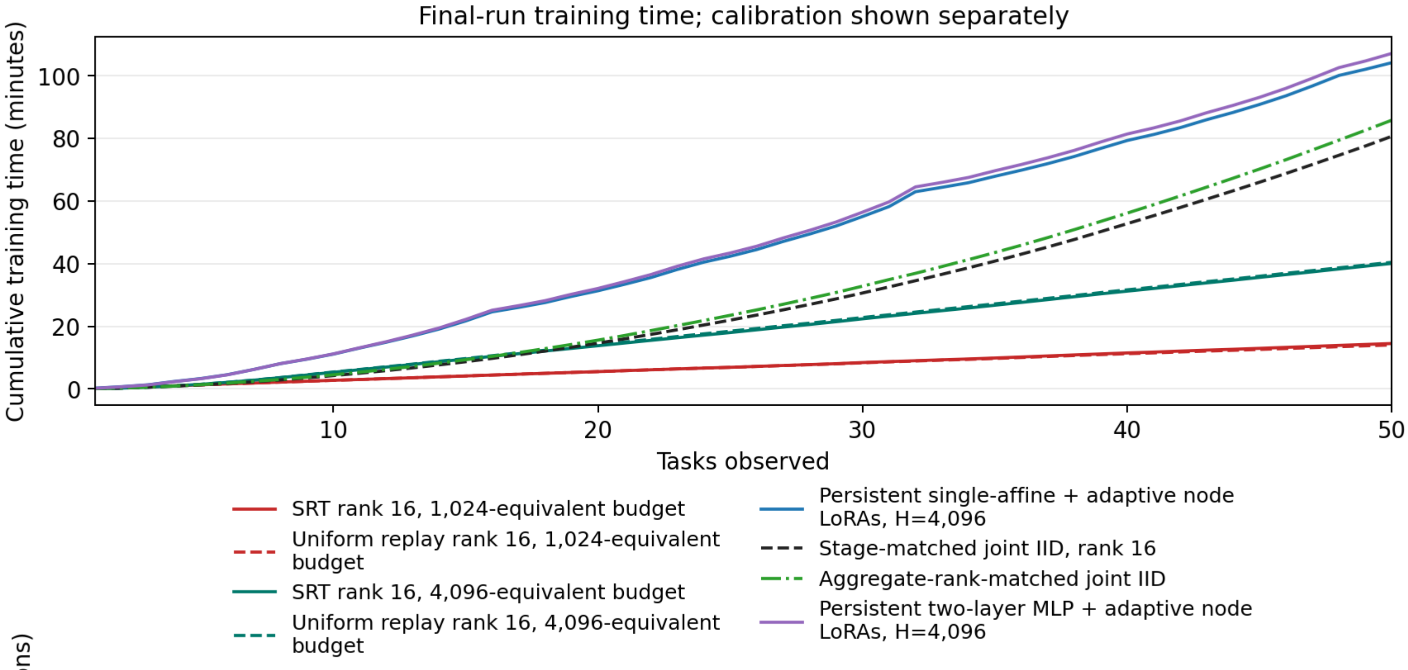
Each new training presentation produces one ViT forward/backward pair. Confidence comes from that same forward: there are no quality-only passes, source hierarchy jobs, or activation recomputation. Prior frontier costs include their source hierarchy. The table separates their extra recomputation forwards.

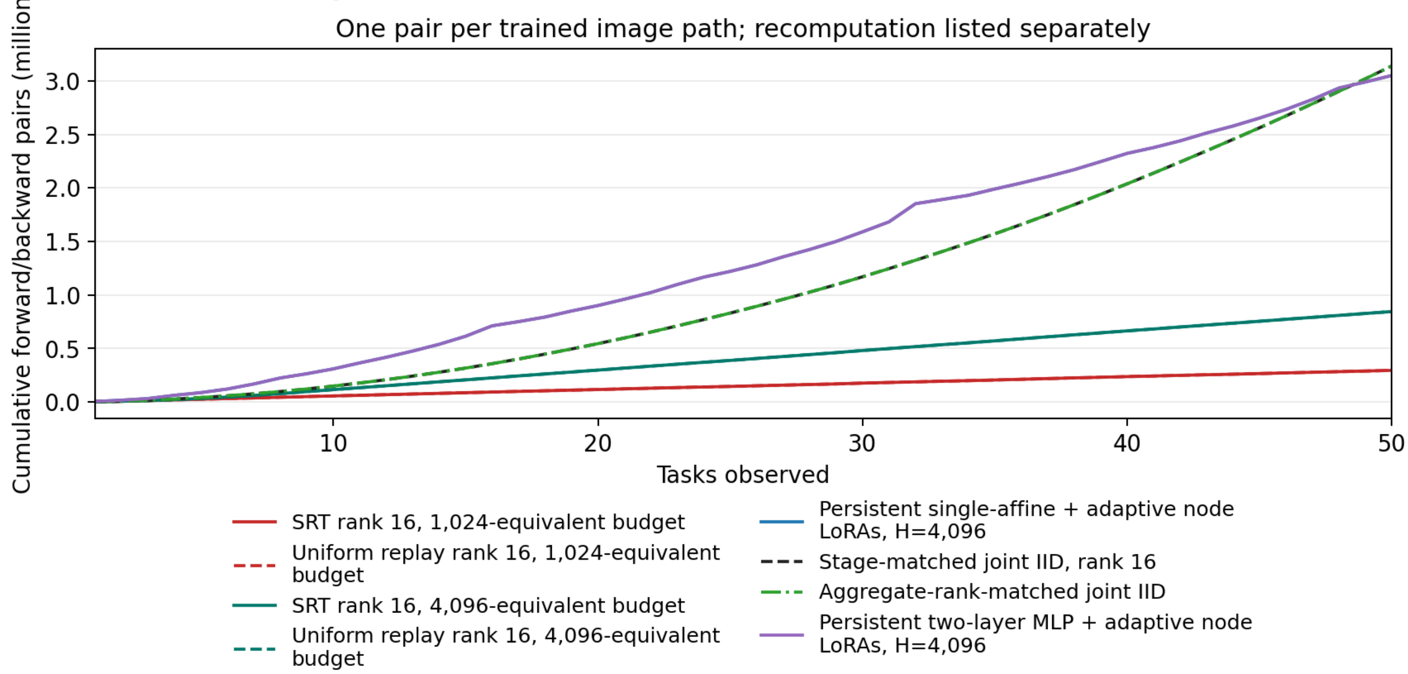
Training time sums measured batch work and committed checkpoint I/O. Loader and scheduler timings are components of that time, not costs to add again. Setup, model serialization outside those counters, between-job overhead, and evaluation are separate. Calibration is excluded from this comparison.

Condition	Forward/backward pairs	Recompute forwards	Train min.
SRT rank 16, 1,024-equivalent budget	294,368	0	14.52
Uniform replay rank 16, 1,024-equivalent budget	294,368	0	14.05
SRT rank 16, 4,096-equivalent budget	844,640	0	40.11
Uniform replay rank 16, 4,096-equivalent budget	844,640	0	40.46
Persistent single-affine + adaptive node LoRAs, H=4,096	3,050,765	2,385,440	104.23
Stage-matched joint IID, rank 16	3,140,210	0	80.72
Aggregate-rank-matched joint IID	3,140,210	0	85.88
Persistent two-layer MLP + adaptive node LoRAs, H=4,096	3,050,765	2,385,440	107.19

Cumulative work across the stream

The fixed H-equivalent budget gives linear training-image work for bounded task size and this fixed model. Repeated prefix testing is quadratic. CPU population scans and report reductions are not covered by the linear model-work claim. These counts are image paths, not profiled FLOPs; equal counts do not imply equal arithmetic for different adapter ranks or multiple frontier models.

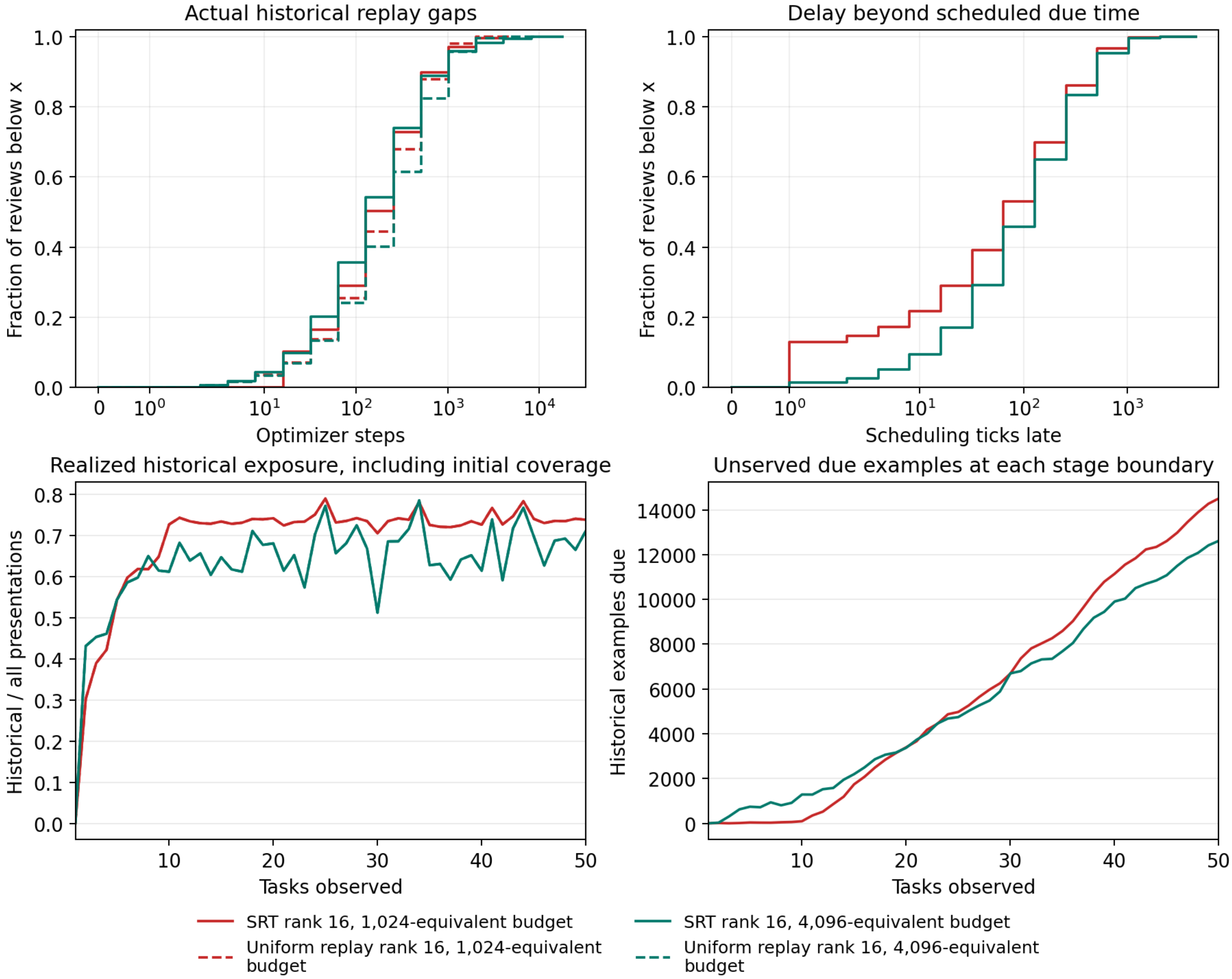




When reviews actually happened

The distributions are review-event weighted and separate real optimizer gaps from virtual scheduling-clock delays. Scheduling time advances to the next due item when neither pool is ready; that does not perform a training update. A large due backlog means requested spacing and actual spacing differ.

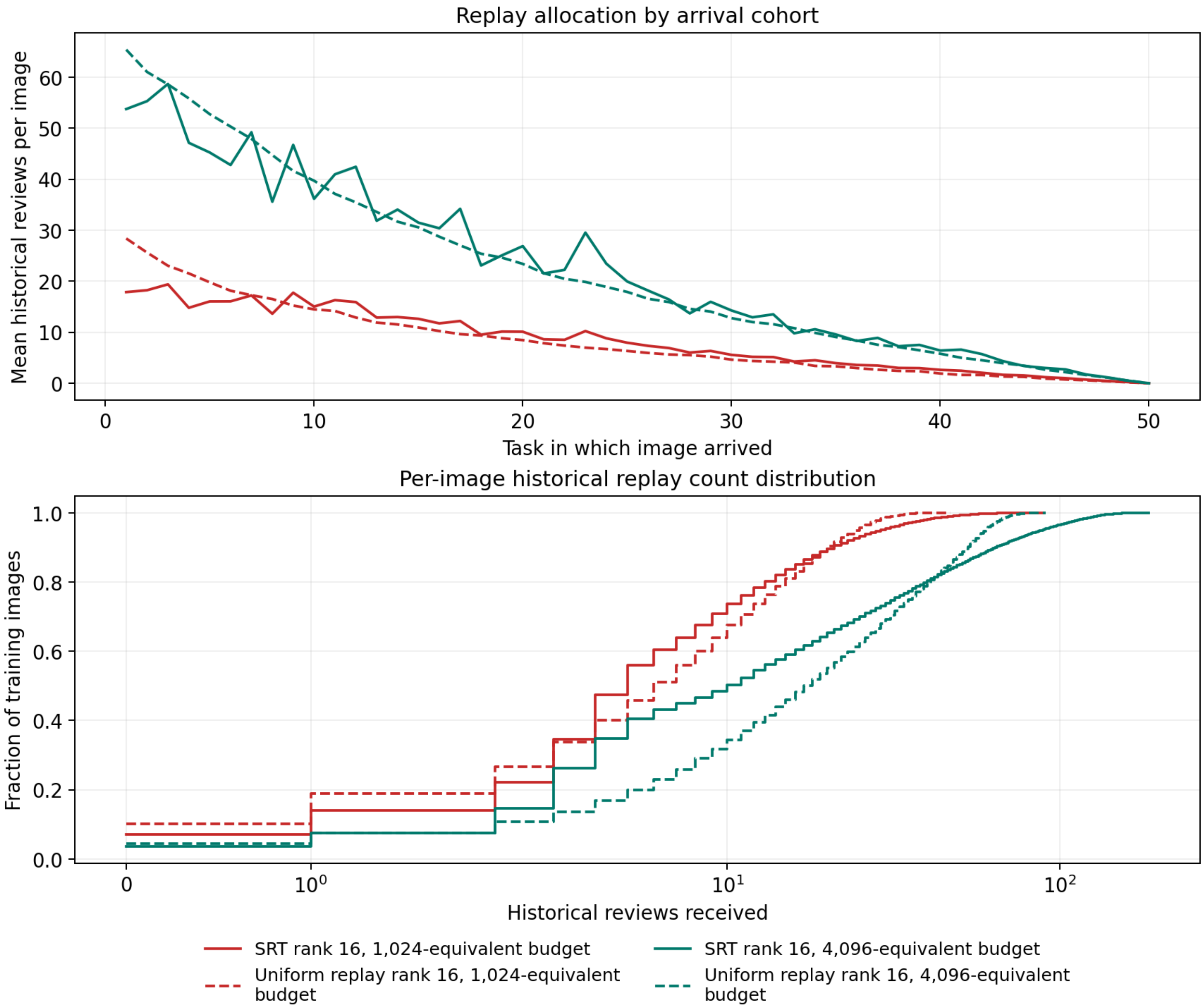
The historical fraction plotted includes the mandatory first pass over new images. Its realized value can differ from the configured post-introduction fraction when one due pool is short. Each uniform control copies the exact resulting old/current counts and partial batch sizes.



Which samples received replay

Earlier arrival cohorts have more opportunities for historical replay. Compare methods within a cohort rather than interpreting that age effect as preferential selection. The lower plot gives each image one vote. Task-50 images cannot receive a historical replay within this experiment.

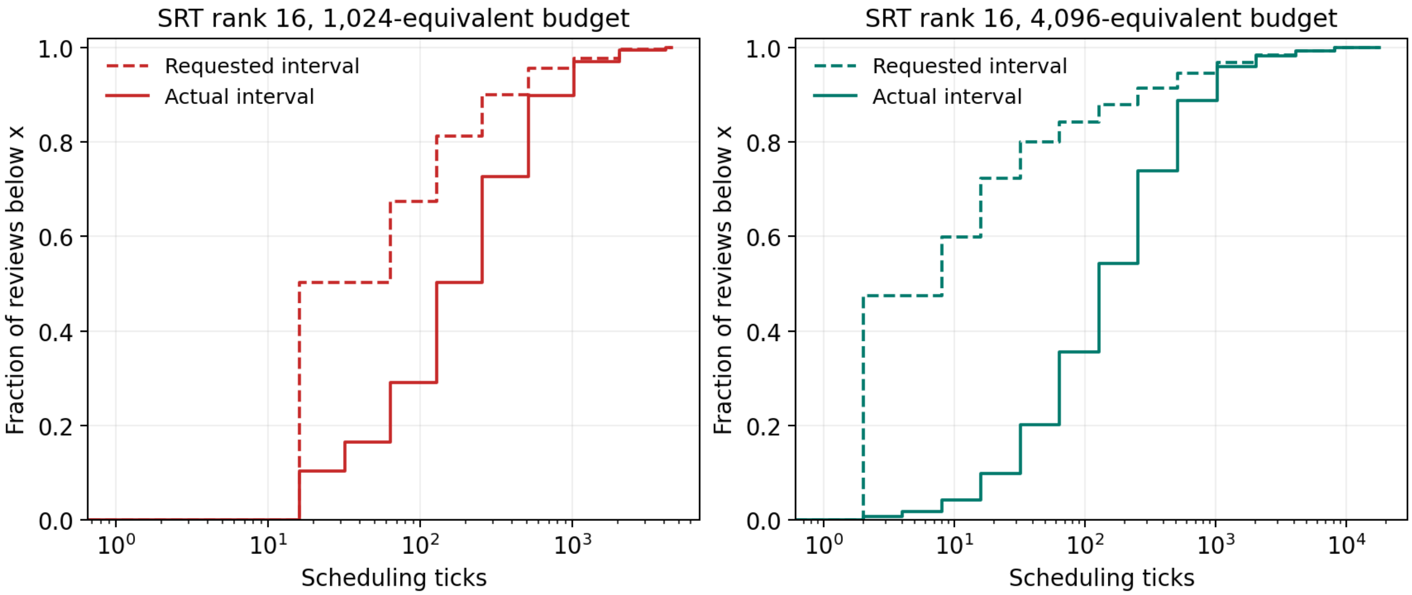
The per-sample table retains exact mean, median, 90th-percentile, minimum, and maximum optimizer gaps, review counts, and terminal waits. Every unfinished next-review interval is marked censored; images with no observed second presentation are retained rather than silently omitted.



Requested spacing versus realized spacing

These curves use the same historical review events and the same scheduling-clock unit. Requested intervals come from the previous SM-2 update; actual intervals run from that previous presentation to the next real review. Any rightward displacement is waiting beyond the requested spacing.

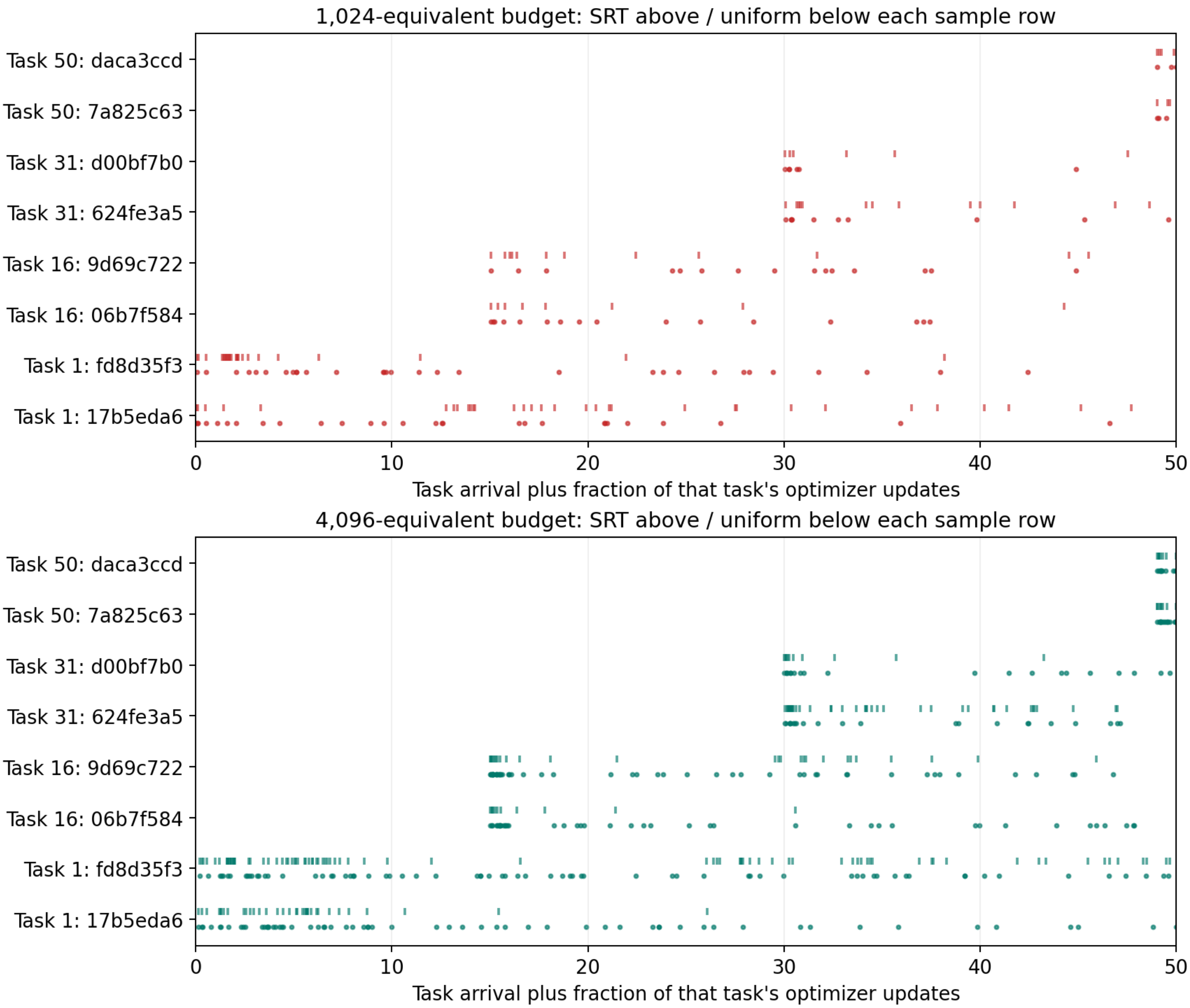
CDF values are evaluated at logarithmic bin boundaries, using completed intervals only. Terminal unfinished intervals are retained separately as censored waits, not interpreted as completed long gaps. The earlier optimizer-gap plot measures actual learning updates rather than virtual clock advances.



Individual review histories

Eight image identities were selected by a fixed hash rule: two each from arrival tasks 1, 16, 31, and 50. Selection does not use accuracy, confidence, or whether SRT looks successful. SRT marks sit above each sample row and uniform marks below. The full identities and every event are retained in the analysis tables.

The horizontal coordinate is task arrival plus the fraction of that task's optimizer updates, not elapsed wall time. Exact optimizer-step gaps are in the timing distributions and per-sample tables.



Calibration and selected settings

Eighteen settings per budget were screened through task 16 on the existing 19,200/4,800 training-derived fit/validation split. Two finalists per budget continued through task 50. Selection maximized mean stage validation accuracy, then minimized mean NLL, then used canonical policy order. No test images entered this process.

Calibration consumed 6,752,992 training presentations, 7.49 measured training hours, and 34.94 evaluation minutes. The four final models restarted from cold adapters after selection.

Relaxed thresholds are [.05,.15,.30,.60,.85], standard [.10,.25,.50,.75,.90], and strict [.20,.40,.70,.85,.95]. Quality counts thresholds met by the pre-update probability of the correct class. The interval unit scales the first/failure interval and the first two successful intervals.

These numerical thresholds were hand-chosen search candidates, not values reported by the paper. Validation selected among them; it did not establish that a quality score predicts retention after a given delay. Calibration costs above exclude the preserved, superseded development run and preflight.

Budget	Threshold profile	Old fraction	Time unit	Mean val. acc.
1,024	standard	0.8	8	82.593%
4,096	strict	0.5	1	80.512%

Complete calibration matrix

Screen accuracy averages tasks 1-16; full accuracy averages tasks 1-50 and is present only for continued finalists. An asterisk marks the selected setting. All values are validation percentages, not test results. NLL, resource counters, hashes, and every stage curve are included in the accompanying tables.

Budget	Profile	Old	Unit	Screen acc.	Full acc.
1,024	relaxed	0.2	1	85.757	-
1,024	relaxed	0.2	8	86.323	-
1,024	relaxed	0.5	1	85.627	-
1,024	relaxed	0.5	8	85.929	-
1,024	relaxed	0.8	1	86.295	-
1,024	relaxed	0.8	8	86.207	-
1,024	standard	0.2	1	84.973	-
1,024	standard	0.2	8	86.578	-
1,024	standard	0.5	1	85.604	-
1,024	standard	0.5	8	87.325	82.475
1,024	standard	0.8	1	86.776	-
1,024	standard *	0.8	8	87.442	82.593
1,024	strict	0.2	1	84.721	-
1,024	strict	0.2	8	86.768	-
1,024	strict	0.5	1	85.732	-
1,024	strict	0.5	8	86.183	-
1,024	strict	0.8	1	87.090	-
1,024	strict	0.8	8	86.743	-
4,096	relaxed	0.2	1	84.917	-
4,096	relaxed	0.2	8	82.927	-
4,096	relaxed	0.5	1	84.462	-
4,096	relaxed	0.5	8	82.950	-
4,096	relaxed	0.8	1	85.071	-
4,096	relaxed	0.8	8	82.946	-
4,096	standard	0.2	1	84.760	-
4,096	standard	0.2	8	84.455	-
4,096	standard	0.5	1	85.222	-
4,096	standard	0.5	8	85.382	-
4,096	standard	0.8	1	85.642	-
4,096	standard	0.8	8	84.835	-
4,096	strict	0.2	1	85.702	-
4,096	strict	0.2	8	85.910	-
4,096	strict *	0.5	1	86.296	80.512
4,096	strict	0.5	8	85.638	-
4,096	strict	0.8	1	86.328	80.226
4,096	strict	0.8	8	85.546	-

Protocol, verification, and interpretation limits

The frozen ViT-B/16 uses rank/alpha-16 LoRA on QKV and fc1 in all twelve blocks. An ordinary affine head adds four rows per task. Old rows, adapter parameters, and SGD momentum persist; all seen rows remain trainable. SGD uses batch limit 64, momentum 0.9, weight decay 0.0005, LoRA learning rate 0.0005, and head learning rate 0.01, without clipping or a learning-rate schedule.

Our implementation follows Atreya et al.'s pre-update scoring and SM-2 equations. Their code is proprietary and their classification thresholds are unspecified. The guaranteed first pass, calibrated thresholds/interval unit, and explicit empty-queue clock rule are documented local choices, not claims of an exact reproduction.

The real smoke test verified zero-LoRA parity, BF16 batch 64, eight task arrivals, exact interrupted/uninterrupted weights and review events, and matched uniform exposure. Reporting revalidates all committed events and reconstructs accuracy/NLL from stored predictions. A completed rerun performs zero optimizer steps.

The uniform control is conditioned on SRT's realized allocation; it is not an independently tuned uniform-replay optimum. Quality is measured on random training crops, so a low score can reflect an uninformative crop as well as forgetting. Hyperparameters were selected offline using a training-derived full-stream validation sweep. One seed cannot establish reproducibility or a publishable state-of-the-art result.

At the final boundary, 14,498 of 24,000 training images are overdue at H=1,024, and 12,560 at H=4,096. Requested spacing is therefore not reliably delivered. This does not isolate the cause of the accuracy difference: crop-dependent confidence, sample selection, and review delays can all contribute.

The next tests should replicate the paired SRT/uniform comparison across seeds and, on training-derived validation data, measure how pre-review confidence predicts later retention at observed delays. Then test spacing calibrated to the available review budget. H changes both work and the selected recipe here, so it is not a pure work ablation. Existing test results must not choose additional settings within this frozen experiment.

Source: Atreya et al. (2026), When to Review: Spaced Repetition for Continual Pre-Training of Language Models, arXiv:2608.17530v1. https://arxiv.org/html/2608.17530v1